



Relative Proximity of Critical Power and Metabolic/Ventilatory Thresholds: Systematic Review and Meta-Analysis

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Abstract

Background Critical power (CP) has been redefined as the new ‘gold standard’ that represents the boundary between the heavy- and severe-exercise intensity domains and hence the maximal metabolic steady state (MMSS). However, several other “thresholds”, for instance, the maximal lactate steady state [MLSS], ventilatory thresholds [VT₁, VT₂] and respiratory compensation point [RCP] have been considered synonymous with CP.

Objective This study aimed to systematically review the scientific literature and perform a meta-analysis to determine the degree of correspondence/difference between CP and MLSS, VT₁, VT₂ and RCP.

Methods A literature search on 2 databases (Scopus and Web of Science) was conducted on October 2, 2019. After analyzing 356 resultant articles, studies were included if they met the following inclusion criteria: (a) studies were randomized controlled trials, (b) studies included interrelations between CP and VT₁, VT₂, MLSS, RCP. Articles were excluded if they constituted duplicate articles or did not meet the inclusion criteria. Nine studies met the inclusion criteria and were included in this meta-analysis. This resulted in 104 participants. A random effects weighted meta-analysis with correlation coefficients was used to pool the results.

Results The pooled correlation coefficient of CP and all thresholds analyzed was $r=0.73$ ($p>0.00001$). The subgroup analysis for each threshold with CP demonstrated significant correlation coefficients of $r=0.80$ (95% CI [0.40; 1.21], $Z=3.90$, $p=0.0001$) for CP & RCP; $r=0.77$ (CI 95% = [0.36; 1.18], $Z=3.71$, $p=0.0002$) for CP & MLSS; $r=0.76$ (CI 95% = [0.31; 1.21], $Z=3.32$, $p=0.0009$) for CP & VT₁. However, CP & VT₂, $r=0.39$ (CI 95% = [-0.37; 1.15], $Z=1.01$, $p=0.31$) were not significantly correlated. Despite the significant correlations between CP and VT₁, MLSS and RCP these variables and VT₂ under- (VT₁, 30%; MLSS, 11%) or over-estimated (RCP, 6%; VT₂, 21%) CP.

Conclusion Regardless of the presence of significant correlations among CP and ventilatory or metabolic thresholds CP differs significantly from each. Thus, logically, if CP represents the best estimate of the heavy-severe exercise intensity transition none of the thresholds considered (i.e., VT₁, VT₂, MLSS, RCP), at least as determined in the studies analyzed herein, should be considered synonymous with such.

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Key Points

Significant correlation coefficients were found between CP and VT₁, MLSS, and RCP.

However, high coefficients of variation were found between CP and the other indices, which supports that the same thing is not being measured.

Thus, CP denotes the heavy-severe exercise intensity transition with relatively little error (SE, 11%), while MLSS will tend to underestimate CP ~ 11% and RCP and VT₂ overestimate CP (by 6 and 21%, respectively).

1 Introduction

For exhausting high-intensity exercise, tolerance increases hyperbolically as a function of decreasing power output [1–3]. This power-time relationship is an important concept in exercise physiology because it provides a systematic framework for understanding the mechanistic bases for fatigue and exhaustion during exercise as well as constituting a tool for fitness diagnostics, training monitoring, and understanding the metabolic responses to exercise [4].

Measurement of the physiological limitations constraining a particular athlete's performance, especially in relation to the power or speed that generates the maximal sustainable oxidative metabolic rate, has proven extremely important for endurance performance athletes [5]. To improve athletic performance coaches design training programs across disparate specific work intensities spanning discrete exercise intensity domains, as defined by their metabolic demands (as dictated by speed, power etc.) and physiologic sequelae, and separated by “thresholds”. Moderate intensity exercise encompasses $\dot{V}O_2$ s below the lactate threshold (LT); here considered synonymous with the ventilatory threshold, VT_1 , especially when detected by gas exchange i.e., $\dot{V}$ -slope [6]. For moderate exercise intensities (i.e., $<LT$, VT_1) and, dependent upon the kinetics of a particular individual, $\dot{V}O_2$ attains its steady-state monoexponentially within ~2–3 min and blood [lactate] remains at resting levels. In contrast, heavy-intensity exercise, above the LT or VT_1 , evidences a secondary, slow component of the $\dot{V}O_2$ kinetics, that delays attainment of a steady-state $\dot{V}O_2$ for perhaps 10–20 min when blood [lactate] also stabilizes at an elevated level. For the highest exercise intensity for which exercise tolerance is sufficient to achieve $\dot{V}O_{2max}$, severe-intensity exercise, it is not possible to achieve a metabolic steady-state (MSS) (ventilation, $\dot{V}O_2$, blood or muscle [lactate] or intracellular pH and high energy phosphates [7, 8]). Presumably, above this critical metabolic rate, there will be different physiological adaptations compared to performing exercise at lower intensities [5, 9].

The boundary between the heavy- and severe-exercise intensity domains has received much attention and, depending on its method of determination and also protocols of determination, has received a plethora of different names [10]. Among these terms are the maximal lactate steady state (MLSS), critical power (CP or speed CS), ventilatory thresholds (VT_1 —a misnomer in this regard—and VT_2) and respiratory compensation point (RCP) [4, 11–21]. Notably, the CP concept has become one of the most widely used parameters for training and performance to evaluate the threshold intensity associated with the lower extremity of the severe-intensity domain [22, 23]. MLSS represents the highest workload that can be maintained without a continuous

blood lactate accumulation during a prolonged submaximal 30-min constant-work rate exercise bout [24]. However, problematically the blood [lactate] at given intensity does not adequately reflect the metabolic status of the muscle [10, 25].

Non-invasive measurements of either the ventilatory profile or the relationship between CO_2 output ($\dot{V}CO_2$) and O_2 uptake ($\dot{V}O_2$, i.e., $\dot{V}$ -slope, denoting the inflection point of $\dot{V}CO_2$ versus $\dot{V}O_2$), typically during a progressive or incremental exercise test, can detect the evolution of the “extra” CO_2 generated by falling bicarbonate levels consequent to buffering of lactic acid-generated H^+ [6] and associated ventilatory measurements (i.e., VT_1) [26]. Moreover, on this type of test the second ventilatory threshold (VT_2), also named the respiratory compensation point (RCP), is identified as the second breakpoint in the ventilation response and corresponds to the end of the so-called isocapnic buffering region where the peripheral chemoreceptors invoke a hyperventilation that drives arterial CO_2 pressure below 40 mmHg [6]. Thus, for exercise in the severe-intensity domain there are at least 3 sources of the CO_2 evolved (i.e., $\dot{V}CO_2$): mitochondrial, bicarbonate-derived, and a hyperventilation-induced lowering of body CO_2 stores [27, 28].

Several studies have investigated the relation of CP with MLSS and VT_2 /RCP [12, 15, 17]. Keir et al. [29] reported that the $\dot{V}O_2$ associated with CP, MLSS and RCP was not different, suggesting that each of these represent the same metabolic rate. However, others have disputed the equivalence of these indexes [11, 20, 21, 30].

Recently, Jones et al. [10] suggested that MLSS and CP may be notionally similar. However, methodological differences in their assessment including: Lack of granularity in the criterion tests and frequency of blood lactate measurements as well as disparities in temporal matching between blood and muscle(s) metabolic profiles can contribute to different power outputs, running speeds and $\dot{V}O_2$ s between MLSS and CP. Those authors provided a strong rationale for considering CP rather than the MLSS—as routinely measured—as the boundary separating heavy- and severe-intensity domains and therefore, the maximal metabolic steady state (MMSS). Because MLSS is typically measured using incrementing work rates or speed the tendency is for MLSS to underestimate MMSS albeit by some modest amount [10]. Specifically, consider that a sequence of constant work rate bouts at 200, 215, 230, 245 and 260 W is chosen to identify MLSS. Suppose that all work rates up to and including 245 W fulfill the lactate steady-state plateau criteria but 260 W evidences a continued lactate increase and thus does not. It can only be said that the MLSS lies somewhere between 245 and 260 W. With respect to VT_1 , as it relates to the initial increase of blood lactate at the moderate-heavy exercise transition rather than that subsequently

at the heavy-severe transition, this variable will substantially underestimate the MLSS [3, 4, 8, 10]. In contrast to VT_1 (and the MLSS), RCP and VT_2 represent secondary sequela to events in the muscle that will be temporally displaced by transit delays to the peripheral chemoreceptors, for example, as necessary to evoke the eventual detected ventilatory responses. Thus, we argue that RCP and VT_2 will occur at greater metabolic rates than the MMSS.

To address the controversy identified above we systematically reviewed the existent scientific literature and conducted a meta-analysis to determine the degree of correspondence/difference between CP and other phenomenon relating purportedly to MMSS such as MLSS, ventilatory thresholds (VT_1 and VT_2) and RCP. Specifically, we tested the hypothesis that MLSS would underestimate the $\dot{V}O_2$, work rate or speed at which CP actually occurs (as would VT_1). Moreover, RCP and VT_2 would necessarily overestimate CP.

2 Methods

This meta-analysis followed the PRISMA statement for improved reporting of meta-analyses [31].

2.1 Search Strategy

A literature search was conducted on October 2, 2019 by two independent reviewers for the following databases: Scopus and Web of Science (Fig. 1). The keywords used to the search were: *critical power, cycling, respiratory compensation point, ventilatory thresholds and maximal lactate steady state*. Abstracts and citations from scientific conferences were excluded.

Title, abstract and keyword search fields were searched using the following search strategy:

Critical power $\times$ AND/OR respiratory compensation point
 $\times$ AND/OR maximal lactate steady state
 $\times$ AND/OR ventilatory threshold $\times$ and cycling.

Searches were limited to human participants and English language only publications or at least the abstract must be in English. Two reviewers (MAG and FGM) performed independently the identification, screening, eligibility and inclusion of studies, with disagreement settled by consensus. All records of literature search were examined by title and abstract to exclude irrelevant records. Studies were selected following the eligibility criteria. Additional records identified through other sources (such as manual searches through article reference lists). Data including the publication details, participant characteristics (recreational or trained cyclists),

testing procedures, study design, description of intervention and results of correlation between CP, MLSS, RCP, VT_1 and VT_2 were extracted from all eligible studies. If insufficient information was reported (e.g. participant characteristics) authors were contacted to confirm additional information about the included studies.

2.2 Inclusion and Exclusion Criteria

Studies published in the English language were included in the present study. This meta-analysis included studies with participants aged from 20 to 32 years and included randomized controlled trials approved by Ethics committees. Data were considered for studies that included interrelations between CP and VT_1 , VT_2 , MLSS, RCP. Articles were excluded if they were duplicate articles and those that did not meet the inclusion criteria mentioned above.

2.3 Data Extraction

A total of 356 studies were identified (Fig. 1). Overall, 43 studies were identified initially that were full-text screened (Fig. 1), but only 9 studies were included in the meta-analysis [12–15, 17–21]. The 34 studies that did not match the eligibility criteria based on full-text screening were discarded because they did not examine the relationship between CP and VT_1 , VT_2 , MLSS and RCP. The studies included were used to extract the relevant parameters and other pertinent variables. Overall, 104 participants aged between 20 and 32 years ($M_{\text{age}} = 24.8$, $SD = 3.45$) were included in the data analysis (Table 1). However, three studies had two comparisons (CP and MLSS, [15], CP and VT_1 [18, 19], CP and VT_2 [19], and CP and RCP [18]. One study [17] had three comparisons (CP and MLSS, VT_1 and VT_2). For this reason, these studies have been listed as “a”, “b” or “c” regarding the comparison.

2.4 Quality Evaluation

The quality of the studies included in the review was evaluated by two observers (MAG and FGM). All the studies being carefully analyzed with the PEDro scale [32] (Table 2). Item 1 is rated for external validity (Yes/No) and items 2–11 for internal validity (rated using 0-absent or 1-present). Given that the assessors are rarely blinded, and that it is impossible to blind the participants and investigators in supervised exercise interventions, the items related to blinding were removed (5–7) from the scale [33]. For this reason, the highest score was 7 points, as the first item is not included in the total score, similar to previous

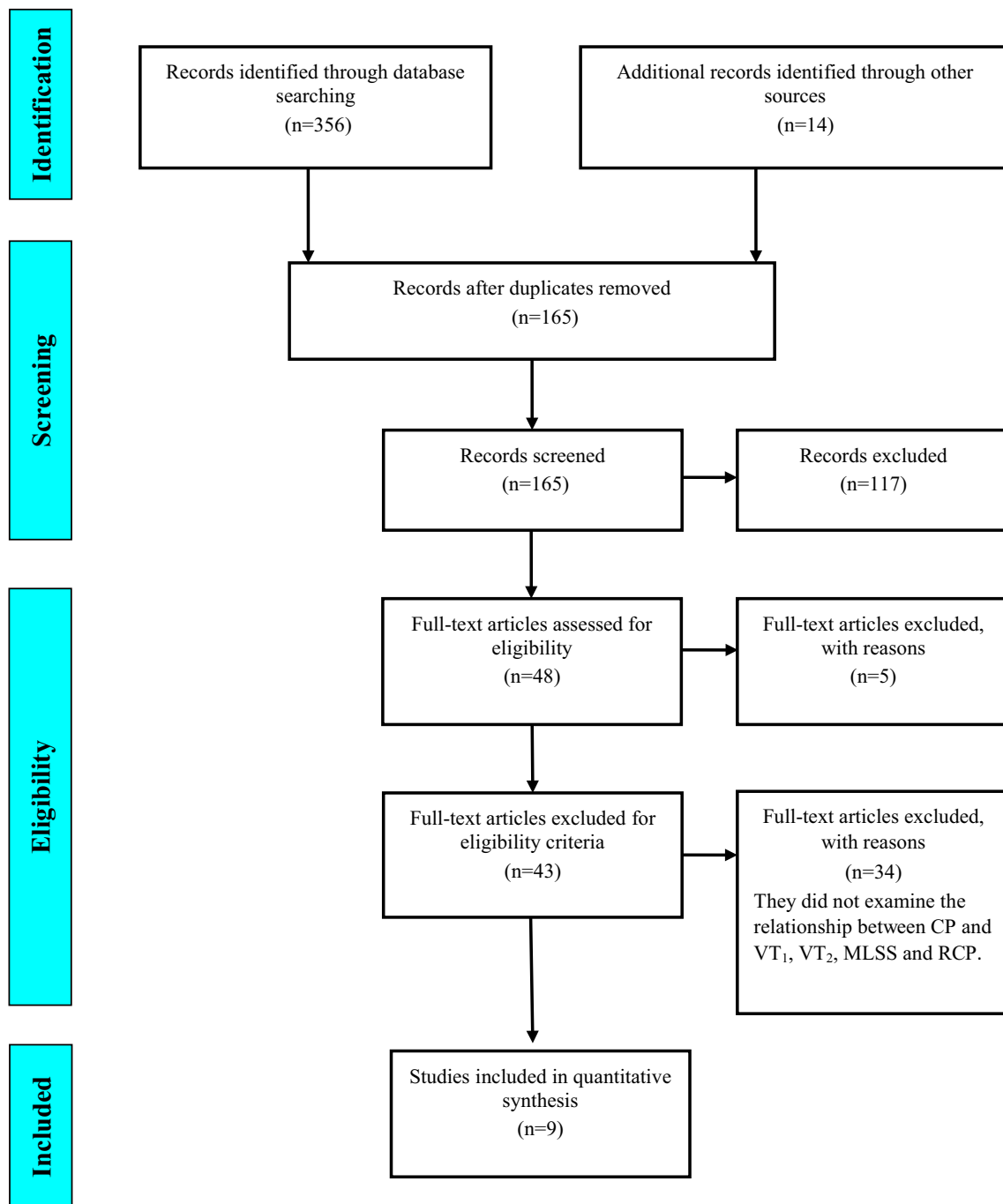


Fig. 1 Flow diagram of the search strategy

systematic reviews [34, 35] as follows: 6–7 = “excellent”; 5 = “good”; 4 = “moderate”; and, 0–3 = “poor”.

2.5 Statistical Considerations

The meta-analysis was performed by using RevMan 5.3 [36]. The statistical method was calculated using inverse variance, using the random effects analysis model for

correlated measures. Heterogeneity of the analyzed studies was assessed using an I^2 -squared (I^2) test. I^2 represents the proportion of effects that are due to heterogeneity as opposed to chance [37, 38]. Thresholds for low, moderate and high levels of heterogeneity correspond to I^2 values of 25, 50, and 75%, respectively. In a subgroup analysis, studies were stratified by metabolic thresholds [16].

Table 1 Studies included in the meta-analysis

Authors	<i>n</i>	Sex (m, f)	Age (mean $\pm$ SD)	Subject	Correlates	
					Association	<i>r</i>
Caritá et al. [13]	6	m	25.5 $\pm$ 4.4 years	Trained cyclists	CP > MLSS	0.99
Greco et al. b [15]	7	m	28 $\pm$ 4 years	Untrained	CP > MLSS	0.98
Mattioni et al. [14]	13	m 9, f 4	26 $\pm$ 3 years	Trained cyclists and Recreational	CP > MLSS	0.95
Pringle and Jones. [12]	8	m 7, f 1	25 $\pm$ 3 years	Trained runners, cyclists, duathletes and recreational athletes	CP > MLSS	0.95
Greco et al. a [15]	7	m	27 $\pm$ 5 years	Trained cyclists	CP > MLSS	0.94
Dekerle et al. a [17]	11	m	23 $\pm$ 2.9 years	Recreational athletes	CP > MLSS	-0.11
Dekerle et al. b [17]	11	m	23 $\pm$ 2.9 years	Recreational athletes	CP > VT ₁	-0.08
Bergstrom et al. a [18]	28	m 14, f 14	23.3 $\pm$ 3.3 years	Recreational athletes	CP > VT ₁	0.81
Nakamura et al. a [19]	11	m	24.4 $\pm$ 3.7 years	Untrained	CP > VT ₁	0.84
Dekerle et al. c [17]	11	m	23 $\pm$ 2.9 years	Recreational athletes	CP < VT ₂	0.07
Nakamura et al. b [19]	11	m	24.4 $\pm$ 3.7 years	Untrained	CP < VT ₂	0.86
Bergstrom et al. b [18]	28	m 14, f 14	23.3 $\pm$ 3.3 years	Untrained	CP < RCP	0.91
Caen et al. [20]	9	m	21.8 $\pm$ 1.2 years	Recreational athletes	CP < RCP	0.69
Leo et al. [21]	11	m	23 $\pm$ 5 years	Recreational athletes	CP < RCP	0.63

VT₁ first ventilatory threshold, VT₂ second ventilatory threshold, MLSS maximal lactate steady state, CP critical power, RCP respiratory compensation point, > the value (W) of CP is higher than the other variable, < the value (W) of CP is lower than the other variable

^{a,b,c}Indicates more than one comparison in the same study

Table 2 PEDro ratings* of the included studies

Study	PEDro ratings											Total
	1	2	3	4	5	6	7	8	9	10	11	
Caritá et al. [13]	Yes	1		1				1	1	1	1	6
Greco et al. [15]	No	1		1				1	1	1	1	6
Mattioni et al. [14]	Yes	1		1				1	1	1	1	6
Pringle and Jones. [12]	Yes	1		1				1	1	1	1	6
Dekerle et al. [17]	Yes	1		1				1	1	1	1	6
Bergstrom et al. [18]	Yes	1						1	1	1	1	5
Caen et al. [20]	Yes	1		1				1	1	1	1	6
Leo et al. [21]	Yes	1		1				1		1	1	5
Nakamura et al. [19]	No	1		1				1		1	1	5

*Items in the PEDro scale: 1=eligibility criteria were specified; 2=subjects were randomly allocated to groups; 3=allocation was concealed; 4=the groups were similar at baseline regarding the most important prognostic indicators; 5=there was blinding of all subjects; 6=there was blinding of all therapists who administered the therapy; 7=there was blinding of all assessors who measured at least one key outcome; 8=measures of at least one key outcome were obtained from more than 85% of the subjects initially allocated to groups; 9=all subjects for whom outcome measures were available received the treatment or control condition as allocated or, where this was not the case, data for at least one key outcome was analysed by "intention to treat"; 10=the results of between-group statistical comparisons are reported for at least one key outcome; 11=the study provides both point measures and measures of variability for at least one key outcome

To assess the variability of the estimations of CP and MLSS, RCP, VT₁ and VT₂, the coefficient of variation (CoV) was calculated.

3 Results

3.1 Level of Evidence and Quality of the Studies

All of the studies selected in the meta-analysis achieved a required standard to be considered to have a low risk of bias (PEDro score ≥ 5 ; Table 2) following the previous systematic reviews [34, 35].

3.2 Characteristics of the Participants

The characteristics of the participants for the studies included in the meta-analysis are shown in Table 1 (total sample size of 104 participants, 19 women). Only three of the studies [12, 14, 18] included men and women, while the rest of studies included only male participants [13, 15, 17, 19–21]. The participants were simply described as recreational athletes in five studies [14, 17, 18, 20, 21], trained cyclists, runners and duathletes in four studies [12–15], untrained subjects in three studies [15, 18, 19] and three studies [14, 15, 18] had two classifications. Five studies assessed the comparison between CP and MLSS [12–15, 17], three studies between CP and VT₁ [17–19], two studies between CP and VT₂ [17, 19] and three studies between CP and RCP [18, 20, 21].

3.3 Characteristics of the Studies Selected

3.3.1 Main Effect

Overall, the pooled correlation coefficient of CP and all metabolic thresholds analyzed (Fig. 2) was $r=0.73$ (95% CI [0.50, 0.96], $Z=6.30$, $p<0.00001$).

3.3.2 Subgroup Analysis

Furthermore, a subgroup analysis of each metabolic/ventilatory threshold with respect to its proximity relative to CP was calculated (Fig. 2). The calculated correlation coefficients were $r=0.80$ (95% CI = [0.40; 1.21], $Z=3.90$, $p=0.0001$) for CP & RCP; CP & MLSS, $r=0.77$ (CI 95% = [0.36; 1.18], $Z=3.71$, $p=0.0002$); CP & VT₁, $r=0.76$ (CI 95% = [0.31; 1.21], $Z=3.32$, $p=0.0009$), and CP & VT₂, $r=0.39$ (CI 95% = [-0.37; 1.15], $Z=1.01$, $p=0.31$).

However, the study of Dekerle et al. [17] showed an opposite pattern (negative correlations) compared with the other studies. When we excluded the study of Dekerle et al. [17] in the CP & MLSS correlation, the correlation

coefficient was $r=0.96$ (CI 95% = [0.51; 1.41], $Z=4.17$, $p=0.0001$). For CP versus VT₁, the correlation coefficient was $r=0.82$ (CI 95% = [0.36; 1.28], $Z=3.47$, $p=0.0005$). Finally, for CP & VT₂, the correlation coefficient was $r=0.86$ (CI 95% = [-0.22; 1.94], $Z=1.56$, $p=0.12$), only with results from the study of Nakamura et al. [19].

VT₁ and MLSS systematically underestimated CP by ~30 and 11%, respectively. In contrast, RCP and VT₂ overestimated CP by 6 and 21%, respectively (Fig. 3).

4 Discussion

This meta-analysis demonstrates that, despite high and significant correlation coefficients between CP and other indices of physiological function, specifically VT₁, MLSS, and RCP, CP represents a unique work rate such that the other ventilatory and metabolic thresholds are unlikely to constitute the maximal metabolic steady-state [MMSS, 3, 4, 7]. Thus, CP denotes the transition between heavy- (sustainable) and severe-intensity exercise where exhaustion occurs in a finite time as predicted by CP and the maximum work capacity performed above CP (W') [3, 4, 38]. Consequently, any differences between CP and, individually, VT₁, MLSS, RCP, or VT₂ likely constitute differential physiological responses (i.e., VT₁) or measurement imprecision based upon either testing methodology (see Jones et al. [10] for MLSS) and/or inherent signaling delays between exercising muscles and generation of the measured output signal (e.g., [lactate], $\dot{V}_E$) [30, 39, 40]. The principal findings herein include: (i) high significant correlations between CP and VT₁ (0.76), CP and MLSS (0.77), as well as CP and RCP (0.80) (Fig. 2). However, CP and VT₂ were not significantly correlated ($r=0.39$, $p=0.31$); (ii) high CoVs were found between CP and the other indices (CoV = 16–22%, Table 3), which supports that, irrespective of the correlations, the same thing is not being measured. (iii) MLSS significantly underestimated CP which has been considered the ‘gold standard’ demarcator of the boundary between heavy- and severe-intensity exercise Jones et al. [10]. However, it should also be kept in mind that there are different ways to measure CP (e.g., different numbers of criterion tests, contrasting modelling approaches, 3-min all-out test) and, thus, it should not be presumed that CP determination as measured in the candidate studies analyzed herein, is error free.

On average, in the studies selected, the power output at CP has been reported to be 11% higher than MLSS [12–15, 17], 6% lower than RCP [18, 20, 21], 30% higher than VT₁ [17–19] and 21% lower than VT₂ [17, 19] (see Fig. 3).

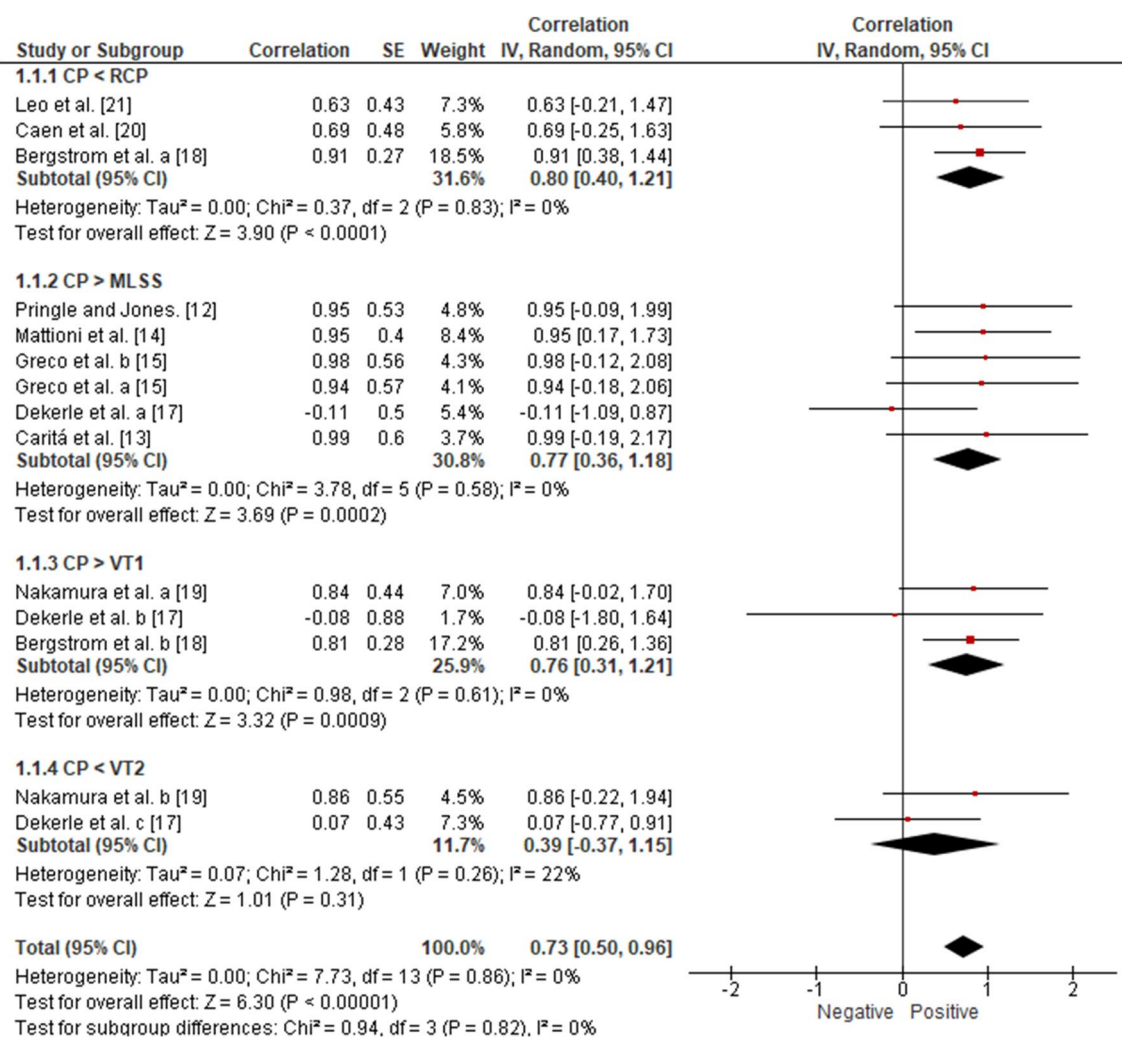


Fig. 2 Forest Plot of the pooled correlation coefficient of CP and Metabolic/Ventilatory Thresholds. Negative correlation, Positive correlation, CI confidence interval, SE standard error, IV weighted correlation

4.1 Placing different “threshold” concepts in perspective

Underlying Physiology:

4.1.1 Ventilation Threshold 1 (VT₁)

As proposed originally by Wasserman and McIlroy [26] and termed the “anaerobic threshold”, VT₁ is measured most conveniently during an incremental or ramp cycle ergometer test. The underlying premise is that, above a certain metabolic demand, lack of sufficient oxygen supply to the exercising muscles necessitates glycolytic ATP production and accumulation of lactic acid. Thus, lactate and the dissociated H⁺ efflux into the blood stream where the H⁺ are buffered by plasma bicarbonate which constrains the fall in arterial pH and produces “extra” CO₂ to be evolved at the mouth via increased ventilation. This response forms the

basis for detection of the ventilation threshold (VT₁, sometimes by the $\dot{V}_E/\dot{V}O_2$ and $\dot{V}_E/\dot{V}CO_2$ ratios and/or end-tidal O₂ and CO₂ pressures) and thus constitutes a non-invasive surrogate for detection of the lactate threshold [40]. Because of the noisiness of the breathing pattern in many subjects and also, perhaps, differences in the sensitivity of the peripheral chemoreceptors among subjects Beaver and colleagues [6] developed the $\dot{V}$ -slope detection method that relies on detecting the breakpoint between VCO₂ and VO₂ thereby removing reliance on ventilatory sensitivity and breathing pattern per se. The term “anaerobic threshold” is now generally considered a misnomer following demonstration that the exercising muscle does not become anoxic (e.g., Honig et al. [41]; Richardson et al. [42]; Voter and Gayeski [43; Gayeski et al. 44], and that the lactate threshold represents simply the disparity in rates of lactate appearance and disappearance [45]). Whereas VT₁ and CP were at first considered synonymous close scrutiny of the original data of Moritani

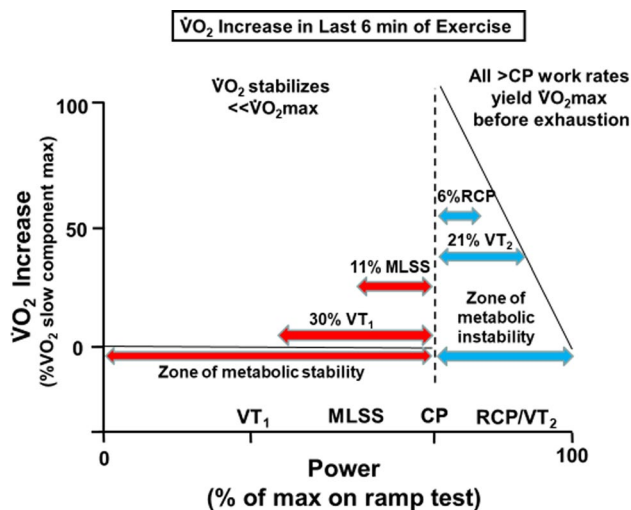


Fig. 3 $\dot{V}O_2$ increase in last 6 min of exercise bouts versus power showing relative proximity of various threshold concepts to Critical Power (CP). VT_1 first ventilatory threshold, VT_2 second ventilatory threshold, $MLSS$ maximal lactate steady state, RCP respiratory compensation point. Please note that the relative proximity of the thresholds, and thus, the power output at which they occur may be protocol specific. $\dot{V}O_{2max}$ maximal rate of oxygen uptake

et al. [46] reveals that the VT_1 occurs typically at ~50–60% $\dot{V}O_{2max}$ compared with 70–80% for CP [3, 47]. Even though the VT_1 – $\dot{V}O_{2max}$ interval typically becomes compressed in highly trained individuals Jones and Poole [48] suggest there will, presumably, always be a heavy intensity domain between VT_1 and CP where blood lactate increases but stabilizes at an elevated level and $\dot{V}O_2$ evinces a slow component that plateaus well below $\dot{V}O_{2max}$ [3, 49]. Regarding the results of the comparison between CP and VT_1 , Nakamura et al. [19] found that the power output associated with CP was between VT_1 and VT_2 , and, therefore, that VT_1 underestimated CP. The studies of Bergstrom et al. [18] and Dekerle et al. [17] also found that VT_1 underestimated CP.

4.1.2 MLSS

MLSS is defined as the highest power output that does not result in a rise of blood lactate greater than 1 mmol/L between 10 and 30 min [24]. The MLSS is typically derived from a series (4–5) of 30 min continuous exercise bouts on separate days, at different but constant power outputs, while blood lactate is measured at rest and each 5 min of exercise [10]. As discussed by Jones et al. [10] the MLSS and CP constitute notionally the same metabolic rate. Both may be determined by means of constant-work rate tests but, unlike CP, the MLSS relies on stepping-up workloads within the heavy-intensity domain to approximate the heavy-severe exercise intensity boundary. The key defining difference rests upon determination of MLSS and, unless an extremely

laborious series of constant-work rate tests—each accompanied by blood lactate determination—is performed, its determination lacks granularity and will always underestimate CP. There are also concerns regarding measurement accuracy, day-to-day variation in blood [lactate] and its response profile due to diet and glycogen depletion as well as the opportunity for unwanted repeated trial effects such as exercise training [10].

Specific to the overall results found herein, in the comparison between CP and MLSS, Mattioni et al. [14] found a correlation coefficient of $r=0.95$, with CP being, on average, 20 W greater than MLSS. Taking into account the wide limits of “agreement” between CP and MLSS, and the discrepancy of power outputs, these authors suggested that CP does not reflect the maximal physiological steady state. In the study of Greco et al. [15] the correlation coefficient was $r=0.94$ – 0.98 , and depended, in part, on the aerobic training status of the subjects (trained vs untrained). However, the difference in power outputs was ~40 W with, again, CP higher than MLSS. In addition, the oxygen uptake associated with CP (~57 ml kg min) was higher than MLSS (~52 ml kg min). These authors concluded that CP and MLSS do not share common putative physiological mechanisms, and also that CP overestimates MLSS by around 4–5 ml/kg/min, whether the subjects are trained or not. In the same vein, Pringle and Jones, [12] found that the power output associated with CP was significantly higher than MLSS, although both variables were strongly correlated ($r=0.95$). The authors suggested that differences of ~20 W and large inter-individual disparities between CP and MLSS indicated that the terms were not necessarily interchangeable. Further to this issue, in the study of Caritá et al. [13] the power output associated with CP was higher than MLSS (315 vs 287 W): This despite the extremely high correlation found ($r=0.99$). Finally, in the study of Dekerle et al. [17], CP and MLSS differed by ~40 W (MLSS being, yet again, higher than CP). However, surprisingly, in this instance CP and MLSS did not correlate ($r=-0.11$). The authors concluded that CP determination leads to an overestimation of the metabolic rate associated with MLSS, and CP and MLSS were assumed to be different physiological parameters.

Thus, on average, the compelling weight of evidence supports that CP is higher than MLSS. As Jones et al. [10] previously discussed, the lack of agreement between MLSS and CP is consistent with the contention that MLSS underestimates the highest steady-state oxidative metabolic rate, and, accordingly, should not be considered the “gold standard” estimate of the MMSS. Also, it is pertinent that, as addressed above, CP itself cannot be resolved in an error-free fashion, with the size of any such error being dependent upon the rigor of its determination.

Table 3 Variability indices of Critical power and Metabolic/Ventilatory Thresholds

Auteurs	CP				MLSS				VT ₂				VT ₁				RCP			
	Mean (w)	SD (±)	SE (±)	CoV (%)	Mean (w)	SD (±)	SE (±)	CoV (%)	Mean (w)	SD (±)	SE (±)	CoV (%)	Mean (w)	SD (±)	SE (±)	CoV (%)	Mean (w)	SD (±)	SE (±)	CoV (%)
Mattioni et al. [14]	253	44	12	17	233	41	11	18	—	—	—	—	—	—	—	—	—	—	—	—
Greco et al. a [15]	325	29	11	9	288	35	13	12	—	—	—	—	—	—	—	—	—	—	—	—
Greco et al. b [15]	202	40	15	20	164	36	14	22	—	—	—	—	—	—	—	—	—	—	—	—
Pringle and Jones. [12]	242	25	9	10	222	23	8	10	—	—	—	—	—	—	—	—	—	—	—	—
Caritá et al. [13]	315	32	13	10	287	38	16	13	—	—	—	—	—	—	—	—	—	—	—	—
Dekerle et al. [17]	278	22	7	8	239	21	6	9	286	28	8	10	159	23	7	14	—	—	—	—
Nakamura et al. [19]	146	33	10	23	—	—	—	—	228	36	11	16	121	28	8	23	—	—	—	—
Bergstrom et al. [18]	187	47	9	25	—	—	—	—	—	—	—	—	145	37	7	26	190	49	9	26
Caen et al. [20]	260	32	11	12	—	—	—	—	—	—	—	—	—	—	—	—	281	28	9	10
Leo et al. [21]	247	43	13	17	—	—	—	—	—	—	—	—	—	—	—	—	268	37	11	14
Mean	246	35	11	15	239	32	11	14	257	32	10	13	142	29	7	21	246	38	10	17

CP critical power, MLSS maximal lactate steady state, RCP respiratory compensation point, VT₁ first ventilatory threshold, VT₂ second ventilatory threshold, SD standard deviation, SE standard error, CoV coefficient of variation

4.1.3 VT_2/RCP

This measurement is also typically determined from an incremental/ramp exercise bout but there are exceptions [50]. Understanding the mechanistic bases for VT_2/RCP is further complicated by the multiple factors that change in a time-dependent fashion during heavy- and especially severe-intensity exercise and may drive increased $\dot{V}_E$. Specifically, for the carotid body ventilatory drive these include: body temperature, blood osmolarity, pH, K^+ , H^+ -buffering by bicarbonate [51] and potentially falling arterial PO_2 in those subjects prone to exercise-induced arterial hypoxemia approaching $\dot{V}O_{2max}$ [52, 53]. There is also an important peripheral ventilatory drive emanating from the greater time-dependent intracellular perturbations (increased K^+ , H^+ and other assorted metabolites) that is driven via Group III/IV afferents [54]. Finally, as fibers fatigue within the active skeletal muscles, additional, less efficient, motor units will be recruited, presumably invoking a greater feed-forward ventilatory drive from the motor cortex. This behavior also adds to the $\dot{V}O_2$ slow component and consequent CO_2 production requiring additional ventilation. These ventilatory requirements are exacerbated further by the falling $PaCO_2$ which mandates a greater $\dot{V}_E/\dot{V}CO_2$ to clear the extant CO_2 load.

The complex etiology of VT_2/RCP , and the challenge of relating either work rate or $\dot{V}O_2$ from ramp-incremental to constant-work rate exercise may well conspire to obscure any relationship with CP—if such a relationship exists. That challenge is substantial as the $\dot{V}O_2$ corresponding to a given work rate at any given time will depend on the underlying kinetics responses, the gain of the primary (fast) and slow components and also specific features related to the imposition parameters (e.g., rate of work increase) of the ramp-incremental test. Notwithstanding this substantial concern it would be most unlikely that the metabolic and gas exchange instability that erupts above the heavy-severe transition (i.e., $> CP$) did not also drive much of the additional $\dot{V}_E$ associated with constant-work rate exercise above VT_2/RCP . That many of the sentinel mechanistic perturbations underpinning VT_2/RCP must be transduced through secondary mediators involving transport-related time-delays and ultimately may depend on the overall sensitivity of ventilatory control it is not surprising that VT_2/RCP in most investigations exceeds CP. Moreover, and especially when an investigation recruits a subject pool with a broad range of $\dot{V}O_{2max}$, the correlation between CP and VT_2/RCP is expected to be high. However, as with any correlation causation is not proven *ipso facto*.

In the common theme of this discussion Nakamura et al. [19] found a correlation coefficient of 0.86 between CP and VT_2 , regardless of the presence of ~ 80 W disparity in their respective work rates. Thus, constant-work rate exercise

at the estimated CP versus VT_2 would necessarily invoke very different times to exhaustion and also time-dependent metabolic and respiratory responses. Somewhat differently, Dekerle et al. [17] found no significant difference in the work rate at CP versus VT_2 but also no correlation. In summary, the evidence supports that, whereas CP and VT_2 do not represent the same physiological phenomenon, they, as well as VT_2 and MLSS may be physiologically related.

In concordance with the concern above regarding extrapolating constant work rates from ramp-incremental tests, in relation to the comparison between CP and RCP, Caen et al. [20] revealed that $CP < RCP$ and they suggested that these terms should not be considered equal, in spite of the high correlations found. Leo et al. [21] reported that the correspondence between CP and RCP is dependent on the protocol selected, as evidenced by the poor measurement agreement (low concordance correlation coefficient ≤ 0.73) and large typical error (≥ 28 W). The mechanistic basis of CP is related to skeletal muscle bioenergetics, substrate utilization and/or motor unit recruitment profile [7, 55]. On the other hand, RCP relies, in part, on the stimulation of peripheral chemoreceptors [56] and the subsequent hyperventilatory response observed during heavy/severe-intensity exercise [57]. Thus, whereas Leo et al. [21] concluded that CP and RCP are unlikely to share the same precise mechanistic origin, the high correlation coefficient ($r=0.91$) between CP (estimated from the 3-min all-out test) and RCP led Bergstrom et al. [18] to postulate a common fatigue mechanism; possibly associated with elevated muscle interstitial and/or arterial $[K^+]$.

4.1.4 Protocol design and limitations

Finally, we must mention some aspects related to the correlation coefficient and protocol design. Despite the correlation coefficients found in the meta-analysis between CP and the other metabolic/ventilatory indices or thresholds, the strength of a correlation depends on the range of the variable(s) of interest in the sampled population. If this range is wide, the correlation will invariably be greater than if it is narrow [58].

Whereas the present findings cannot ratify that CP is the “gold standard” for estimating the MMSS (see 10) and thus, the heavy-severe exercise intensity boundary, they do demonstrate that, if this is indeed the case, VT_1 , MLSS, RCP and VT_2 cannot be considered synonymous with MMSS. However, to reduce the error of estimate of CP within and among investigations some rigorous criteria must be agreed upon and adhered to, amongst which is a consistent testing method. As long ago as 1988 Poole and colleagues recommended using 4 or 5 exhausting constant-work rate tests that spanned supra-CP work rates that induced exhaustion in 2–15 min, approximately [3]. Limiting these individual

bouts to very short durations (i.e., <5–6 min) has been demonstrated to overestimate CP leading to some investigators estimating CP at > 500 watts in a group of endurance runners! [59] see also Poole, [60]. It is also important to state the standard error of the CP estimate in Results in each instance. Moreover it is apparent that, for some data sets, use of different linear versus non-linear models may produce contrasting CP parameter estimates [61]. Given the presence of, at least some, ineradicable experimental error, it is important that investigators cease trying to exercise subjects exactly at CP. This procedure is statistically and physiologically poor science. Specifically, even if the errors follow a Gaussian distribution, tasking them to exercise precisely at the determined CP 50% of the subjects will be above CP, in the severe-intensity domain, and 50% below CP, in the heavy intensity domain. Lastly, use of the 3-min all-out test as pioneered by Vanhatalo, Doust and Burnley [62–65] holds great promise as an expeditious method for determining CP (and W') as well as critical speed (CS) and D' (the curvature constant of the speed-time relationship) [66]. It is certainly deserving of more widespread evaluation across different candidate populations.

Another factor to consider in the comparison between CP and the other metabolic parameters is the reproducibility and reliability of the measurement. Previous studies [67] have found a high test–retest reliability ($r=0.98$) for ventilatory threshold (VT), and fatigue threshold (i.e., CP, Gaesser and Wilson [68]), when each subject performed two pretraining tests. However, the studies selected in this meta-analysis did not measure the reliability of the measurements by means of test–retest procedures. We consider it to be a major strength to incorporate this measure of reproducibility to verify the reliability of the metabolic/ventilatory thresholds or parameters studied.

5 Conclusions

A wealth of intramuscular and whole-body physiological responses supports that CP denotes the heavy-severe exercise intensity transition [10] and also that, depending on the precise determination methodology and modelling approach, CP can be determined with relatively little error (SE, 11%, Table 3). In marked contrast, VT_1 denotes the transition between moderate and heavy-intensity exercise and is therefore expected to, and does, occur at a far lower work rate and $\dot{V}O_2$. MLSS, VT_2 and RCP may all be secondary manifestations of CP with their method of determination as regards ramp versus constant-work rate test and also the method of detection biasing the proximity to CP. Specifically, depending on the rigor and granularity of the testing and measurement protocol, MLSS must necessarily tend to underestimate CP (~11% in the studies analyzed herein) and, considering

that CP is indeed the MMSS, if CP occurs at a different work rate, MLSS cannot be synonymous with the MMSS. Alternatively, where the $\dot{V}_E$ response is the measurement of choice (i.e., RCP, VT_2), complexities of secondary responses to the deteriorating intramuscular and humoral environment coupled with transit/development delays and the receptor sensitivity and response characteristics, including substantial noise, of for example, the ventilatory control system, must be appreciated. It is not surprising, therefore, that RCP and VT_2 overestimate CP (by 6 and 21%, respectively).

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Compliance with Ethical Standards

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